



DEVICE INTEGRATION GUIDE



LoRaWAN™ Operated Sensors

(only for firmware version 6.0.0 and above)

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Device Integration Guide

(LoRaWAN)



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General View of Ellenex LoRaWAN Products

Sensor Type	Series	Standard model	Corrosion resistant	With built-in temperature	With GPS	IECEX Approved	IP68 rated	Other models
Industrial Pressure	PT	PTS2	PTC2	PTD2	PTG2	PTSX	PTS3	PTF2 PTE2
Differential Pressure	PD	PDS2	-	-	PDG2	PDSX	-	PDT2
Submersible level	PL	PLS2	PLC2	PLD2	PLG2	PLSX	PLS3 PLV3	PLM2 PLMD2
Industrial temperature	TT	TTS2	-	N/A	TTG2	TTSX	TTS3	-
Distance (Ultrasonic/ Radar)	D	DUS2	DUC2	-	DUG2	-	DUS3 DUC3	DRC3

Flow	F	FTS2 Standard water flowmeter	FMS2 Industrial Water meter interface	-	-	-
Water Quality	C	CSD2 Conductivity and salinity	CTR2 Turbidity	CPH2 pH	CDO2 Dissolved Oxygen	-
Load/ Strain	L	LTS2 S-Type load cell	LSS2 Foil strain gauge	-	-	-
Moisture/ Rain/ humidity	M	MWS2 wall moisture	MSS2 soil moisture	MAS2 Humidity	MRS2 Rain sensor	
Others	-	GTS3 Geo-location	ODS2 Object detection	ECP2 Cathodic Protection	-	-

Sensor Interface	R	RS1	RM1	RM4	RE1	-
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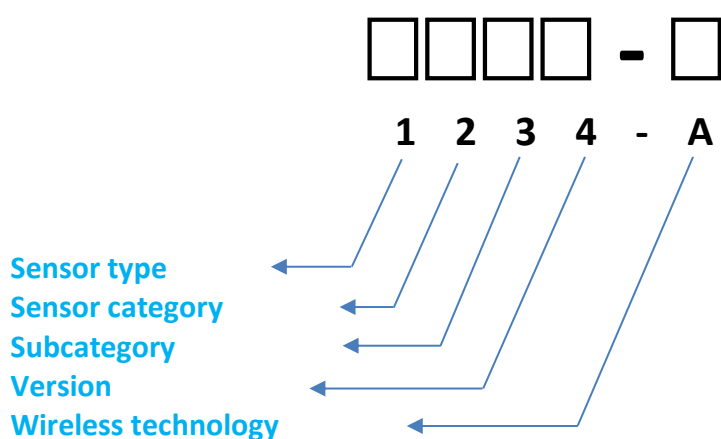
		Single channel sensor interface	Multi-channel sensor interface	4-channel identical sensor interface	Preconfigured multi-sensor	
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Note:

- Products in grey colour are under development and are not commercialised yet.

Guide To Find the Right Product

Ellenex product codes follow a standard format to make it easy to choose. Product codes for our LPWAN Wireless Sensors comes with below format:



Notation	Description	Codes
1 and 2	Sensor Type and Sensor Category	P: Pressure PT: Industrial Pressure PD: Differential Pressure PL: Submersible level T: Temperature D: Distance/ Displacement DU: Ultrasonic DR: Radar F: Flow C: Chemical parameters (mainly for water quality) M: Moisture/ Rain/ humidity L: Load, Strain and force E: Electrical R: Sensor interface
3	Sub-category	S: Standard F: Flush type M: Miniature C: Corrosion resistant D: With built-in temperature sensor G: With built-in GPS

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4	Version	From 1 to 9 or alphabets 2: Generation 2 with built-in features, mainly IP66 rated 3: Mainly IP68 rated devices 4: Ultra-compact and modular design
A	Wireless technology/ Communication	N: NB IoT/ CAT M1 L: LoRaWAN S: Satellite W: Wirepas

Device Technical Summary

LoRaWAN Specification Version	1.0.2
Supported Class	Class-A
Supported Frequency Plans	AS-923, US-915, EU-868, AU-915, and other frequencies upon request
ADR	Supported and enabled by default
Confirmation	Disabled by default and can also be enabled by default.
RF Range (distance)	Many variables determine range, >15K is possible
Antenna Type	Sleeve dipole, omni directional
Antenna Gain	Approx. 2dB
Default TX Power (prior to successful join)	Max TX permitted by region.
Maximum TX Power	Depends on the regional frequency allowance, it is either +16 dBm or +20 dBm
Maximum Enabled Channels	In device standard configuration, 8 channels are enables, but more channels can be enabled by MAC commands, based on the published "Notice of Use and Disclosure" by LoRa Alliance
Control Interface	There are OTA control commands
Default Uplink Interval	4 hours but can be changed by a downlink command
Trigger LED via Downlink	All Downlinks to device trigger the LED on the device.

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Device Commissioning Instructions (OTAA Device)

If you are not using ellenex LoRaWAN network server, follow the below instruction:

First, make sure device is switched off, then enter the device keys in the corresponding LoRaWAN network server. Turn on the device by pressing the on/off switch (latching switch) or placing the on/off jumper in place (if device ordered without on/off switch. The LED around the key should start blinking instantly. If the device successfully connects to the network the LED will blink again 4 times followed by one extra blink (1, 4, 1 blinks). If you didn't see any more blinks, it means the trial for connection has not been successful. To trouble shoot, investigate connection issues by following steps below:

- Make sure your gateway has internet connectivity and is correctly added to the network server you are trying to add the device.
- Make sure the device keys are correct, and the device is added to the network server that your gateway is connected to.
- Check the frequency plan of the gateway including the sub-bands and adjust it to the frequency plan of the device if it is different.

After successfully connecting the device to the network, you should see device data in your network server or visualization platform. You may observe 3-5 additional short-interval transmissions from the device. Then, the device switches to the standard sampling rate. The standard sampling rate of devices is generally 4 hours, but you can change it to any sampling rate allowed by your network by sending a downlink to the device.



When you change the sampling rate, please be careful to provide a correct sampling rate to avoid draining the battery due to an unnecessarily fast sampling rate.



After each power cycle, you should wait at least 30 seconds to power on the device again.

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Tags Description

Ca	Current Amplitude in Amperes (A)
Cf	Current Frequency in Hertz (Hz)
Cm	Current Median in Amperes (A)
Con	Conductivity in micro-Siemens per centimetre ($\mu\text{S}/\text{cm}$)
D	Distance in metres (m)
DO1	Dissolved Oxygen in Saturation Percentage (%Sat)
DO2	Dissolved Oxygen in milligram per Liter (mg/l)
DO3	Dissolved Oxygen in parts per million (ppm)
DP	Differential pressure in bar (bar) for PDS sensors
DP	Differential pressure in Pascals (Pa) for PDT sensors
L	Level in meters (m)
mA	Raw milliamperes in milliamperes (mA)
mV	Raw millivolts reading in millivolts (mV)
P	Pressure in bar (bar)

pH1	pH in pH (pH)
pH2	pH in millivolts (mV)
ps	State of the sensor for sensors with external power (EPB option)
Pu	Pulse (Number of pulse count)
Red	Redox
Res	Resistivity in Ohm meters (Ωm)
RH	Relative Humidity in percentage (%RH)
Sal	Salinity in parts per thousand (ppt)
T	Temperature in degree Celsius ($^{\circ}\text{C}$)
TR1	Turbidity in Nephelometric Turbidity Unit. (NTU)
TR2	Turbidity in Formazin Turbidity unit (FNU)
TR3	Turbidity in milligram per Liter (mg/l)
V	Charge Level in Volts (V)
Va	Voltage Amplitude in Volts (V)
Vf	Voltage Frequency in Hertz (Hz)
Vm	Voltage Median in Volts (V)

Payload Format

Ellenex used **CBOR** (Concise Binary Object Representation) in the new generation of device firmware. It provides more flexibility for more complex devices and addresses limitations of LoRaWAN protocol for multi-sensor integrations.

What is CBOR?

CBOR is a data format designed for small code size, small message size, and extensibility without the need for version negotiation. It is a binary format that allows the encoding of data structures like maps, arrays, and simple data types such as integers and strings. Unlike text-based formats such as JSON, CBOR's binary nature reduces the size of the encoded data significantly. This reduction in size is especially beneficial in environments where bandwidth and data storage are limited, such as in LoRaWAN industrial applications.

Why Ellenex Uses CBOR?

Ellenex employs CBOR for sensor data decoding due to several compelling reasons:

- **Compactness:** The binary format of CBOR ensures that data is encoded in a highly compact manner, which is crucial for LoRaWAN communication where packet size is limited. Smaller data sizes mean that more information can be transmitted within the constraints of LoRaWAN's payload limits.
- **Efficiency:** CBOR's design allows for fast and efficient encoding and decoding operations. This efficiency is vital for Ellenex sensors, which are often battery-powered and need to perform their tasks using minimal computational resources. Efficient data processing may help to extend the battery life of these sensors, enabling longer deployment periods without maintenance.
- **Flexibility:** and more important, CBOR supports a wide array of data types and complex data structures, enabling Ellenex to encode various sensor readings and configurations within a single packet. This flexibility is critical as it allows different combinations of complex sensor data, to be included in one message format without altering the overall data structure.
- **Interoperability and Standardisation:** CBOR is a standardised format, ensuring compatibility with various systems and applications. This standardisation facilitates easier integration and data processing on the receiving end, allowing for seamless data analysis and reporting in different software environments. New data structure of ellenex devices provides possibility of offering solutions for very complex industrial applications, while providing very simple data feed of final calibrated reading of device to the network server and the software platform. It ensures reliable and efficient data transmission, helping Ellenex deliver robust IoT solutions that meet the demanding requirements of modern industrial applications.

CBOR LoRaWAN Data Packets

LoRaWAN packets have size constraints depending on the data rate being used. After joining a network, the packet size may be more conservative, resulting in packets being split into smaller sizes. Typically, there is a lower limit of 51 bytes per packet, providing flexibility in data transmission.

Key-Value Pairs: The parameters in LoRaWAN packets are formatted as key-value pairs. A key-value pair is a data type comprising two pieces of data: a key (identifier) and a value (associated data).

- **Key:** A constant used to define the data set.
- **Value:** A variable belonging to the data set.

In essence, a key-value pair functions like an array, mapping keys to their corresponding values. This format is extensively used in databases to identify data uniquely, ensuring no repetitions of the same key.

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Payload Size Considerations: For LoRaWAN, the worst-case scenario for payload sizes is 11byte for the single packet data transmission and generally 51 bytes, assuming no MAC responses are queued. This allows the CBOR format to manage data efficiently.

Example Payload Configuration:

- **6 Sensor Readings:** Utilising the available 51 bytes, the payload can include data from up to six different sensors.
- **5 Sensor Readings + Battery Voltage:** Alternatively, the payload can include data from five sensors and the battery voltage.

Tag and Result Size: It is possible to break down the CBOR packet as follows:

- A byte to mark the start of the array.
- Each string (tag) takes at least $n+1$ bytes. For a 2-byte tag, it will be 3 bytes.
- Each float takes 5 bytes.
- A byte at the end to mark the end of the array.

For n measurements, the packet size s can be calculated as:

$$s=2+n*(3+5)$$

Where:

- 2 accounts for the start and end bytes.
- 3 includes the overhead for the string (tag) for each measurement.
- 5 is the size of the float for each measurement.
- n is the number of measurements.

Following table illustrates the packet size for each number of measurements.

Number of readings	Packet Size
1	10
2	18
3	26
4	34
5	42
6	50

How to decode CBOR?

CBOR is a binary data format that can be decoded to retrieve the original data. Decoding CBOR data involves either using platform-specific payload decoders or manually decoding using online tools. By understanding the structure of CBOR packets and following the appropriate steps, you can efficiently retrieve and interpret sensor data from ellenex LoRaWAN transmissions.

Decoding CBOR data can be achieved through the following methods:

1. Using a Payload Decoder Provided by the Platform Most IoT platforms, such as The Things Network (TTN), provide built-in payload decoders for CBOR data. These decoders are tailored to the specific data structure and automatically parse the CBOR data into human-readable JSON format. Users need to select or configure the appropriate decoder within their platform's interface to enable automatic decoding of received data packets.

2. Manual Decoding Steps If a platform does not provide a built-in decoder or custom decoding is required, CBOR data can be manually decoded by following these steps:

- **Identify and Capture the Payload:** Start by capturing the payload from the received LoRaWAN packet. The payload is typically provided in a hexadecimal format.
- **Split the Payload:** After every power cycle, the first transmission may contain two packets due to the inclusion of MAC commands. These packets need to be identified and split for decoding.

Example of Decoding CBOR Data:

Let's consider the following two packets captured after a power cycle:

Packet 1:

BF625661FA4068A3D7FF = {"Va": 3.634999990463257}

Packet 2:

BF62566DFABDBE76C8625666FA42FA0000624361FA403820C462436DFA00000000624366FA430EDB656176190C52FF = {"Vm": -0.09299999475479126, "Vf": 125.0, "Ca": 2.876999855041504, "Cm": 0.0, "Cf": 142.8570098876953, "v": 3154}

To decode these packets, you can use an online CBOR decoder such as cbor.me. Follow these steps:

1. **Copy the Hexadecimal String:** Copy the hexadecimal string from the payload you want to decode.
2. **Paste into the Right-Hand Side of Decoder:** Go to cbor.me and paste the hexadecimal string into the input field. (Note: Input field is on the right, and Output field is on the left).
3. **Decode:** Click on the green arrow in the input field. The tool will automatically decode the CBOR data and display the corresponding JSON format.

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Decoded Output:

For Packet 1:

Hexadecimal: BF625661FA4068A3D7FF

JSON: {"Va": 3.634999990463257}

For Packet 2:

Hexadecimal: BF62566DFABDBE76C8625666FA42FA0000624361FA403820C462436DFA00000000624366FA430EDB656176190C52FF

JSON: {"Vm": -0.09299999475479126, "Vf": 125.0, "Ca": 2.876999855041504, "Cm": 0.0, "Cf": 142.8570098876953, "v": 3154}

CBOR conversion on <https://cbor.me/>

CBOR

Diagnostic ☒ plain hex ☐ deterministic (with ☐ dCBOR)

47 Bytes ☐ as text ☐ utf8 ☐ emb cbor ☐ cborseq
enter hex below or No file chosen

```
{ "Vm": -0.09299999475479126, "Vf": 125.0, "Ca": 2.876999855041504, "Cm": 0.0, "Cf": 142.8570098876953, "v": 3154 }
```

```
BF      # map(*)
62      # text(2)
566D    # "Vm"
FA BDBE76C8 # primitive(3183376072)
62      # text(2)
5666    # "Vf"
FA 42FA0000 # primitive(1123680256)
62      # text(2)
4361    # "Ca"
FA 403820C4 # primitive(1077420228)
62      # text(2)
436D    # "Cm"
FA 00000000 # primitive(0)
62      # text(2)
4366    # "Cf"
FA 430EDB65 # primitive(1125047141)
61      # text(1)
76      # "v"
19 0C52    # unsigned(3154)
FF      # primitive(*)
```



To maintain the flexibility and integratability of the data format, Ellenex uses floating-point numbers (floats) for sensor readings. Floats provide the flexibility to provide optimised resolution based on the device range and specs and better representation of the real reading from the sensing element, which is critical factor for precise monitoring solutions. However, this precision requires careful handling, particularly when it comes to truncating or rounding decimal places based on the specific sensor's measurement range and accuracy level. These details are outlined in the product datasheet and must be considered when configuring the decoder or software platform. By tailoring the decimal precision to match the sensor's capabilities and application requirements (for most of applications, 2 or 3 decimals), Ellenex ensures that the transmitted data is both accurate and compatible with various systems and use cases.

Payload Decoder

Link: [Version 6 Payload decoder.](#)

This link contains the payload decoder for CBOR packets.

This decoder is for all the devices and decodes data as per the LoRa WAN Integration guide. This has not been implemented on TTN Repository. If the device is Version 6 then this decoder has to be used.

This decoder must be manually added to the device payload formatter on any platform. Steps to add it to platform. Example for TTN.

1. Log in to the TTN Console: Visit the TTN Console and log in with your account credentials.
2. Add the device to the preferred Application: Once logged in, go to Applications from the sidebar and add the device and go to that device.
3. Go to Payload Formatters: After going into the device section, navigate to the Payload Formatters tab.
4. Choose Uplink Formatters: Under the Payload Formatters section, you'll see options for both Uplink and Downlink formatters. Choose Uplink to decode data being sent from your device to the network.
5. Select Custom JavaScript: In the Uplink Payload Formatter section, select Custom JavaScript. This allows you to write custom JavaScript to decode your device's payload.
6. Write the Decoder Function: Copy the whole code from cbor.js file and paste it.
7. Save the Decoder: After Pasting the decoder function, click Save Changes to apply the decoder to the application.
8. Monitor Data: Once the decoder is in place, it will automatically decode the uplink messages from your device. You can check the Data tab in the application to see decoded payloads.



Please be mindful to use this decoder for devices with the Version 6 firmware.

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Standard Payload Formats

Sensor Category	Sensor type	Parameter	Unit	Conversion Example
Pressure	PTS2-L	- Pressure (P) - Battery Charge Level (v)	bar mV	BF6150FA3FD8A5C86176190CF4FF
	PTC2-L			<pre>{"P": 1.6925592422485352, "v": 3316}</pre>
	PTS3-L			- Pressure = 1.692bar - Battery Charge Level = v/1000 = 3.316V
	PTF2-L			
	PTD2-L	- Pressure (P) - Temperature (T) - Battery Charge Level (v)	bar °C mV	BF6150FA3FA9ED1C6154FA41D47AE16176190CFE <pre>{"P": 1.3275485038757324, "T": 26.559999465942383, "v": 3326}</pre> - Pressure = 1.327bar - Temperature = 26.55°C - Battery Charge Level = v/1000 = 3.326V
	PTE2-L	- Pressure (P) - Battery Charge Level (v)	kg/cm ² mV	BF6150FA3FD8A5C86176190CF4FF <pre>{"P": 1.6925592422485352, "v": 3316}</pre> - Pressure = 1.692 kg/cm² - Battery Charge Level = v/1000 = 3.316V
Differential Pressure	PDS2-L	- Differential Pressure (DP) - Battery Charge Level (v)	bar mV	BF624450FA3F4EAEC96176190CF8F <pre>{"DP": 0.8073545098304749, "v": 3320}</pre> - Differential Pressure = 0.807bar - Battery Charge Level = v/1000 = 3.320V

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	PDT2-L (standard range)	- Differential Pressure (DP) - Temperature (T) - Battery Charge Level (v)	Pa °C mV	BF624450FA3B8888896154FA41C00 0006176190C28FF {"DP": 0.004166666883975267, "T": 24.0, "v": 3112} - Differential Pressure = 0.004Pa - Temperature = 24.0°C - Battery Charge Level = v/1000 = 3.112V
	PDT2-L (customised range -100 to 3500 Pa)	- Raw Voltage - Battery Charge Level (v)	mV mV	BF61561901026176190C2AFF {"V": 258, "v": 3114} - Raw Voltage = 258mV - Battery Charge Level = v/1000 = 3.114V Formula: Differential Pressure (Pa) = Ifactor*(Raw voltage-250)/3750 Ifactor=3500 Pa Here, Differential Pressure (DP) = 3500*(258-250)/3750 = 7.467 Pa
Level	PLS2-L	- Level (L) - Battery Charge Level (v)	m mV	BF614CFA3FCEC8C86176190CF8FF {"L": 1.6155023574829102, "v": 3320} - Level = 1.6155023574829102m - Battery Charge Level = v/1000 = 3.320V
	PLC2-L			
	PLM2-L			
	PLS3-L	- Primary Level (L1) - Primary Pressure (P1) - Battery Charge Level (v) - Total Level (L)	m bar mV m	BF625031FAB8E4B700624C31FA400 181F66176190D04FF {"P1": -0.00010905973613262177, "L1": 2.023557186126709, "v": 3332} Total Level(L) = L1-(P1*10) = 2.021m
	PLV3-L			
	PLC3-L			

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				Battery Charge Level = $v/1000 = 3.320V$
	PLD2-L			BF614CFA3FF20AFA6154FA41DECCC D6176190CF4FF
	PLMD2-L	- Level (L) - Temperature (T) - Battery Charge Level (v)	m °C mV	{"L": 1.8909599781036377, "T": 27.850000381469727, "v": 3316} - Level = 1.890m - Temperature = 27.85°C - Battery Charge Level = $v/1000 = 3.316V$
Temperature	TTS2-L			BF6154FA41BD30356176190C34FF
	TTS3-L	- Temperature (T) - Battery Charge Level (v)	°C mV	{"T": 23.64853858947754, "v": 3124} - Temperature = 23.648°C - Battery Charge Level = $v/1000 = 3.124V$
Distance (Ultrasonic)	DUS2-L			BF6144FA3FF7EF9E6176190D06FF
	DUS3-L			{"D": 1.937000036239624, "v": 3334}
	DUC2-L	- Distance (D) - Battery Charge Level (v)	m mV	- Distance = 1.937m - Battery Charge Level = $v/1000 = 3.124V$
	DUC3-L			
Distance (Radar)	DRC3-L	- Distance (D) - Battery Charge Level (v)	m mV	BF6144FA3FF7EF9E6176190D06FF
				{"D": 1.937000036239624, "v": 3334}
				- Distance = 1.937m - Battery Charge Level = $v/1000 = 3.124V$
Flow Interface	FMS2-L	- Pulse Count (Pu) - Battery Charge Level (v)	- mV	BF6250750C6176190C34FF
				{"Pu": 12, "v": 3124}
				Pulse = 12

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				- Flow= Pu* Pulse per Liter - Battery Charge Level = v/1000 = 3.124V
Water Quality	CSD2-L	- Conductivity (Con) - Salinity (Sal) - Resistivity (Res) - Temperature (T) - Battery Charge Level (v)	μS/cm ppt Ωm °C mV	BF6154FA41BAE35363436F6EFA424203126353616CFA3CD42C3C63526573FA41C3DF3B6176190C1CFF {"T": 23.360998153686523, "Con": 48.50299835205078, "Sal": 0.02589999884366989, "Res": 24.483999252319336, "v": 3100} - Conductivity = 48.503μS/cm - Salinity = 0.025ppt - Resistivity = 24.483Ωm - Temperature = 23.361°C - Battery Charge Level = v/1000 = 3.100V
	CTR2-L	- Turbidity (TR1) - Turbidity (TR2) - Turbidity (TR3) - Temperature (T) - Battery Charge Level (v)	NTU FNU mg/l °C mV	BF6154FA41C5AE1563545231FA40DEC49C63545232FA40DEC49C63545233FA4043851F6176190D04FF {"T": 24.71000099182129, "TR1": 6.96150016784668, "TR2": 6.96150016784668, "TR3": 3.055000066757202, "v": 3332} - Turbidity = 6.961NTU - Turbidity = 6.961FNU - Turbidity = 3.055mg/l - Temperature = 24.71°C - Battery Charge Level = v/1000 = 3.332V
	CPH2-L	- pH(pH1) - Redox (Red) - pH(pH2) - Temperature (T) - Battery Charge Level (v)	pH mV mV °C mV	BF6154FA41BC26EA63704831FA40DE816F63526564FAC10BF97263704832FA40999A6C6176190C1EFF {"T": 23.519001007080078, "pH1": 6.9532999992370605, "Red": -8.74839973449707, "pH2": 4.800100326538086, "v": 3102}

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				- pH = 6.953pH - Redox = 8.748mV - pH = 8.748mV - Temperature = 23.519°C - Battery Charge Level = v/1000 = 3.102 V
	CDO2-L	- Dissolved Oxygen (DO1) - Dissolved Oxygen (DO2) - Dissolved Oxygen (DO3) - Temperature (T) - Battery Charge Level (v)	%Sat mg/l ppm °C mV	BF6154FA41C0D4FE63444F31FA3FAE90FF63444F32FA3A37803363444F33FA3F303AFC6176190D6EFF {"T": 24.104000091552734, "DO1": 1.3637999296188354, "DO2": 0.0006999999168328941, "DO3": 0.6884000301361084, "v": 3438} - Dissolved Oxygen = 1.363%Sat - Dissolved Oxygen = 0.0007mg/l - Dissolved Oxygen = 0.688ppm - Temperature = 24.104°C - Battery Charge Level = v/1000 = 3.438
Soil Moisture	MSS2-L			Based on the customer's requirement.
Humidity	MAS2-L	- Relative Humidity (RH) - Temperature (T) - Battery Charge Level (v)	% °C mV	BF625248FA420300006154FA41C2A3D76176190D12FF {"RH": 32.75, "T": 24.329999923706055, "v": 3346} - Relative Humidity = 32.75 %RH - Temperature = 24.329999923706055 °C - Battery Charge Level = v/1000 = 3.346 V
Rain Sensor (Tipping Bucket)	MRS2-L	- Pulse Count (Pu) - Battery Charge Level (v)	- mV	BF6250750C6176190C34FF {"Pu": 12, "v": 3124} Formula Rain = Pu * Rain rate

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				- Battery Charge Level = $v/1000 = 3.124V$ Example: - Rain rate = 0.2mm - Here, Rain = $12 \times 0.2 = 2.4mm$
Cathodic Protection	ECP2-L	- Voltage Amplitude (Va) - Voltage Median (Vm) - Voltage Frequency (Vf) - Current Amplitude (Ca) - Current Median (Cm) - Current Frequency (Cf) - Battery Charge Level (v)	V V Hz A A Hz mV	BF625661FA4068831262566DFABD BE76C8625666FA42B5D169624361F A3F0B020D62436DFA000000006243 66FA430EDB656176190DCEFF {"Va": 3.632999897003174, "Vm": -0.09299999475479126, "Vf": 90.90900421142578, "Ca": 0.5430000424385071, "Cm": 0.0, "Cf": 142.8570098876953, "v": 3534} - Voltage Amplitude = 3.633V - Voltage Median = 0.093V - Voltage Frequency = 90.909Hz - Current Amplitude = 0.543A - Current Median = 0.0A - Current Frequency = 142.857Hz - Battery Charge Level = $v/1000 = 3.534V$
Single Channel Interface	RS1-L	Pulse: - Pulse count (Pu) - Battery Charge Level (v)	- mV	BF6250750C6176190C34FF {"Pu": 12, "v": 3124} - Pulse = 12 - Battery Charge Level = $v/1000 = 3.124 V$
		4-20 mA: - Raw mA Reading(mA) - Battery Charge Level (v)	microA mV	BF626D41190F596176190C2EFF {"mA": 3929, "v": 3118} - Raw reading = $3929/1000 = 3.929mA$ - Battery Charge Level = $v/1000 = 3.118V$

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		0-10V and 0-5V: - Raw Voltage (V) - Battery Charge Level (v)	V mV	BF61561901026176190C2AFF {"V": 258, "v": 3114} - Raw Voltage = $V/1000 = 0.258\text{ V}$ - Battery Charge Level = $v/1000 = 3.114\text{ V}$
		mV: - Raw Voltage (mV) - Battery Charge Level (v)	μV mV	BF626D561901986176190D7CFF {"mV": 408, "v": 3452} - Raw reading = $408/1000=0.408\text{mV}$ - Battery Charge Level = $v/1000 = 3.114\text{V}$
		Pt-100 or Pt1000: - Temperature (T) - Battery Charge Level (v)	$^{\circ}\text{C}$ mV	BF6154FA41BD30356176190C34FF {"T": 23.64853858947754, "v": 3124} - Temperature = 23.648°C - Battery Charge Level = $v/1000 = 3.124\text{V}$
		RS232 RS485 I2C SDI-12 Modbus		Based on Customers' Requirements.
Multi-Channel Interface	RM1-L	Pulse+ mV+ Pt 100+ 4-20mA: - Raw Voltage (mV) - Temperature (T) - Raw mA Reading (mA) - Pulse count (Pu) - Battery Charge Level (v)	μV $^{\circ}\text{C}$ μA - mV	BF626D56397AB46154FA41BD7106 626D41190F59625075106176190CE CFF {"mV": -31413, "T": 23.680187225341797, "mA": 3929, "Pu": 16, "v": 3308} - Raw mV reading = -31413 mV - Temperature = 23.680°C - Raw mA reading = -3929 mA - Pulse = 16 - Battery Charge Level = $v/1000 = 3.308\text{V}$
		Pulse + Pt 100 + mV: - Raw Voltage (mV) - Temperature (T) - Pulse count (Pu) - Battery Charge Level (v)	μV $^{\circ}\text{C}$ - mV	
		Pt100 + 4-20mA: - Temperature (T) - Raw mA Reading (mA) - Battery Charge Level (v)	$^{\circ}\text{C}$ μA mV	

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		Pulse + 4-20mA: - Raw mA Reading (mA) - Pulse count (Pu) - Battery Charge Level (v)	μA - mV	
		Pulse+ Pt 100 +4-20mA: - Temperature (T) - Raw mA Reading (mA) - Pulse count (Pu) - Battery Charge Level (v)	$^{\circ}\text{C}$ μA - mV	
		Pulse +mV: - Raw Voltage (mV) - Pulse count (Pu) - Battery Charge Level (v)	μV - mV	
		Other sensor combinations		Based on Customers' Requirements.
	RM4-L	4 Channel 4-20mA - Raw mA Reading 1(mA1) - Raw mA Reading 2(mA2) - Raw mA Reading 3(mA3) - Raw mA Reading 4(mA4) - Battery Charge Level (v)	μA μA μA μA mV	BF636D4131190F52636D4132190F7F636D4133190F91636D4134190FA6627073016176190D8EFF {"mA1": 3922, "mA2": 3967, "mA3": 3985, "mA4": 4006, "ps": 1, "v": 3470} - Raw mA reading(mA1) = $3922/1000 = 3.922\text{mA}$ - Raw mA Reading 2(mA2) = $3967/1000 = 3.967\text{mA}$ - Raw mA Reading 3(mA3) = $3985/1000 = 3.985\text{mA}$ - Raw mA Reading 4(mA4) = $4006/1000 = 4.006\text{mA}$ - Battery Charge Level = $v/1000 = 3.410\text{V}$
	RE1-L			Based on Customers' Requirements.
Devices with External power and battery backup (EPB)	EPB	- ps = 0 (Device is operating on external power supply) - ps = 1 (Device is operating on battery)		BF626470FABC0888896174FA41C000006176190DC4FF {"dp": -0.0083333333767950535, "t": 24.0, "ps": 0, "v": 3524}
Devices with Dry Contact	Dry Contact	- DC=0(Open Circuit). - DC=1(Closed Circuit, Device wakes up and transmits.		BF624443016176190D90FF {"DC": 1, "v": 3472}

Downlinks

HOW TO CHANGE DEVICE SAMPLING RATE

After each power cycle, the devices send 10 initial samples every 5 minutes for troubleshooting, then it reverts to its default sampling rate (4 hours).

To change sampling rate, the "sleep period" needs to be changed by sending a downlink data to port 5 using the format 10YYXXXX:

- YY: 00 for seconds, 01 for minutes
- XXXX: hexadecimal value of seconds/minutes you want to set
- For example, setting sleep period to 180 minutes (3 hours), the command will be 100100B4, while for setting to 45 seconds, it will be 1000002D. Currently, minimum sleep period is 60 seconds. Contact support if periods below this is required,

After an OTA command is received and correctly recognized by the end device, it will echo its value back. The payload of the message echoed back will comprise a hex number 0xAAAABBCCCCDD interpreted as follows:

- AAAA - The board serial number
- BB - 0x01 - the type of message - 0x01 indicates a configuration event message
- CCCC - The command that was sent
- DD - The battery voltage in tenths of a volt

Please be careful when sending OTA commands to the device. Incorrect commands may change sensitive properties which may result in the device malfunction.

HOW TO ENABLE/DISABLE CONFIRMATION VIA DOWNLINK

To switch confirmations on, send message data 0x0701

To switch confirmations off, send message data 0x0700

Over the air commands must be sent to Port 5.

After an OTA command is received and correctly recognised by an end device, it will echo its value back. The payload of the message echoed back will comprise a hex number 0xAAAABBCCCCDD interpreted as follows:

- AAAA - The board serial number
- BB - 0x01 - the type of message - 0x01 indicates a configuration event message
- CCCC - 0x0700 or 0x0701 - The command that was sent
- DD - The battery voltage in tenths of a volt

Please be careful when sending OTA commands to the device. Incorrect commands may change sensitive properties which may result in the device malfunction.

HOW TO RESET THE DEVICE VIA DOWNLINK

If the end-device is remote and needs to be restarted there is an OTA downlink command to do so: 0xFF00. Sending this to port 5 will reset the device.

Please be careful when sending OTA commands to the device. Incorrect commands may change sensitive properties which may result in the device malfunction.

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Changing counters and constants via downlink. (Only for the devices using counters and constants in conditional code.)

Packet starts with command byte 0x06 (counters) and 0x08 (constants), then has one or more blocks of the format `ii aabbccdd`

Where:

- `ii` is the index of the counter to change (0...7)
- `aa` is the MSB and `dd` is the LSB for the 32-bit number to be used.



Note that the counters/constants are signed, so anything over 0x80000000 will be interpreted as a negative number.

More than one counter/constant can be set with this command (if the packet size allows it) so the format could be:

```
06 iiaabbccdd (Counter)
08 iiaabbccdd (Constant)
```

The response for this packet is `06 iiaabbccdd` or `08 iiaabbccdd` – Device will respond to the packet same as the downlink command.



Please be mindful that we can change all the counters/constants at one by using the above format but because of the limitation of LORAWAN, here we suggest you change the counter/constant one by one as US915 has 51 bytes only, it might chop off the remaining bytes and the command will not be executed.

Examples of changing first counter.

Downlink command

```
06 00 00 00 00 0a
```

- 06= Command byte to change the counters
- 00= Counter0
- 00 00 00 0a= Changing the counter0 to 10 pascals (Remember it must be converted to hexadecimal)

The response from the device will be

```
06 00 00 00 00 0a
```

Examples of changing first constant.

Downlink command

```
08 00 00 00 00 0a
```

- 08= Command byte to change the constants.
- 00= Constant0

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- 00 00 00 0a= Changing the counter0 to 10 pascals (Remember it must be converted to hexadecimal)

The response from the device will be

08 00 00 00 00 0a

Technical Support

For any technical support or question about customised data format, please contact our technical support team: SUPPORT@ELLENEX.COM

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Integrated IoT Solutions

E:sales@ellenex.com

W: www.ellenex.com